

Report

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1. My Implementation

1.1 Load the provided stereo image pair

The pipeline begins by loading left and right stereo images using the **Python Imaging Library (PIL)**. Both images are converted into **NumPy arrays** to enable pixel-level manipulation.

1.2 VR Lens Distortion

To simulate the optical characteristics of VR headset lenses, I implemented barrel distortion using the **Brown–Conrady radial distortion model**:

$$r' = r (1 + k_1 r^2 + k_2 r^4)$$

For each pixel in the output image:

Coordinates were normalized relative to the image center. The distortion formula was applied to compute the source location in the input image. Nearest-neighbor sampling was used to assign pixel values, with boundary checks to avoid out-of-range access.

This pre-distorts the image so that when viewed through curved VR lenses, straight lines appear correctly aligned to the user.

1.3 Chromatic Aberration

Chromatic aberration arises because lenses refract different wavelengths at slightly different angles. I simulated this by scaling each RGB channel independently around the image center:

1.4 Foveated Rendering

Red channel: scaled outward by 1% ($r_scale = 1.01$)

Green channel: unchanged ($g_scale = 1.0$)

Blue channel: scaled inward by 1% ($b_scale = 0.99$)

For each pixel, new sampling coordinates were computed per channel using:

$$x' = (x - cx) * scale + cx$$

$$y' = (y - cy) * scale + cy$$

Human vision has high acuity only in the central foveal region. To utilize this, I implemented **distance-dependent Gaussian blur** centered at the gaze point.

Step:

- ① Computed Euclidean distance from every pixel to the gaze point.
- ② Defined two radii: Inner radius (100 px) and Outer radius (250 px)
- ③ For pixels between these radii, blur strength was interpolated linearly.

2. Parameter Choices and their Effects

Effect	Parameter	Value	Visual Impact
Lens Distortion	<code>k1</code>	0.25	Strong barrel curvature; corrects pincushion from lenses
	<code>k2</code>	0.15	Adds higher-order distortion for realism
Chromatic Aberration	<code>r_scale</code>	1.01	Red fringes outward at edges
	<code>b_scale</code>	0.99	Blue fringes inward
Foveated Rendering	<code>inner_radius</code>	100 px	Central region remains sharp
	<code>outer_radius</code>	250 px	Blur ramps up smoothly beyond this
	<code>max_sigma</code>	10	Strong peripheral blur; simulates low-acuity vision

3. Visual Effect



Figure 1 Lens Distortion

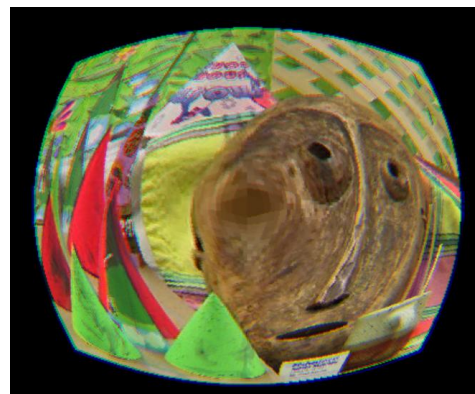


Figure 2 Chromatic Aberration



Figure 3 Foveated Rendering

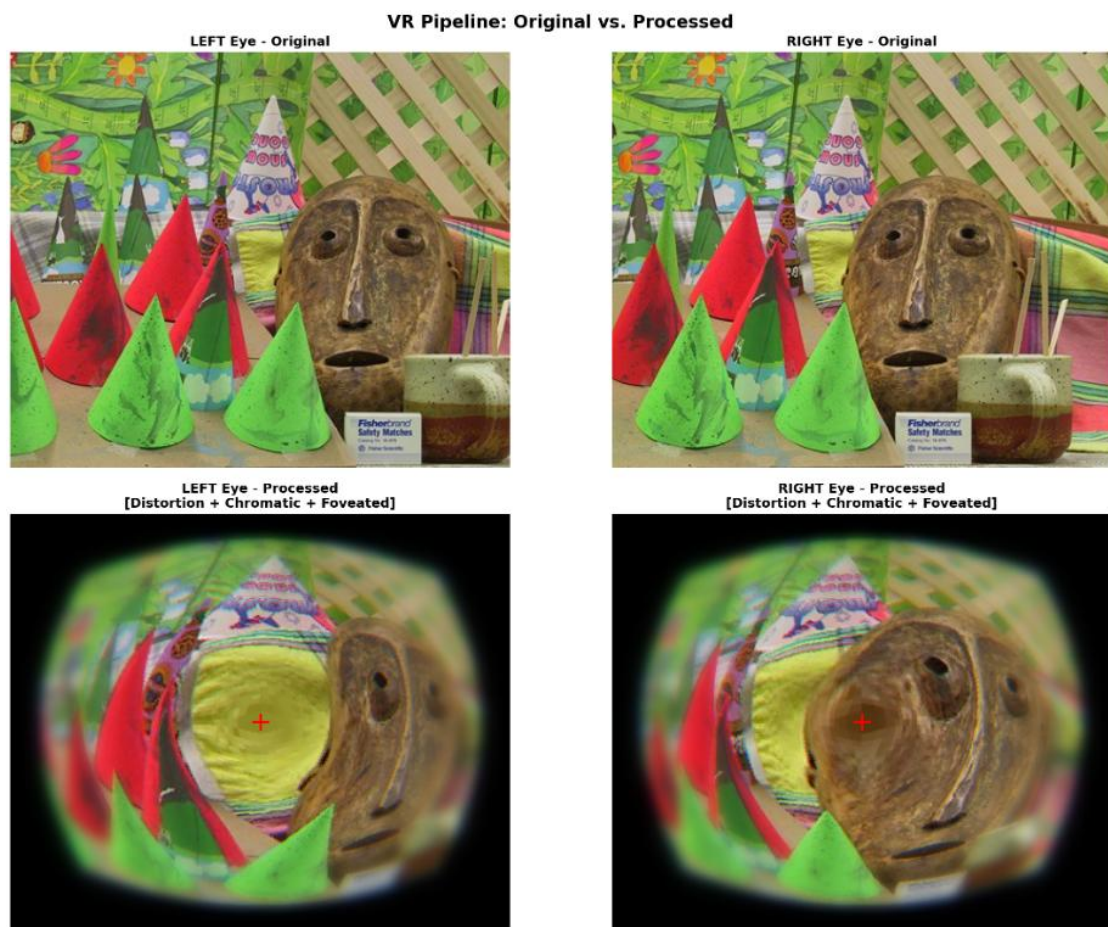


Figure 4 VR Pipeline